RAMAKRISHNA MISSION VIDYAMANDIRA

(Residential Autonomous College affiliated to University of Calcutta)

B.A./B.Sc. SIXTH SEMESTER EXAMINATION, MAY 2025

THIRD YEAR [BATCH 2022-25]

COMPUTER SCIENCE [Honours]

Paper : DSE 3

Date : 10/05/2025

Time : 11 am – 1 pm

Full Marks : 50

Answer **any five** questions of the following : [5×10]

1. a) You are given purchase patterns from a retail store. Apply k-means clustering (k = 3) on the data and interpret the result.

C-ID	Visit per month	Average card value (Rs.)
C1	1	1200
C2	7	700
C3	2	1000
C4	5	2000
C5	3	1500
C6	4	1400
C7	6	1000
C8	5	1200

b) Explain how Federated Learning differs from traditional centralized machine learning. Why is Federated Learning considered privacy-preserving? [7+3]

2. a) Use logistic regression to predict whether a person will subscribe for LIC. Estimate the coefficients using gradient descent (one epoch is enough).

Person	Age	Income per month (₹)	Subscribed
A	30	50,000	Yes
B	38	60,000	Yes
C	25	30,000	No
D	40	60,000	Yes
E	28	35,000	No

b) Define cost function and its role in logistic regression. [7+3]

3. a) You're provided with a sensor dataset from a greenhouse. Apply Decision Tree logic to classify if water is needed (Yes or No)

Record	Soil Moisture (%)	Temp (°C)	Humidity (%)	Water Needed
1	35	30	70	No
2	20	35	60	Yes
3	40	28	75	No
4	18	32	50	Yes

b) Mention two real-world applications of decision trees beyond agriculture. [7+3]

4. a) The following data shows equipment conditions and whether a failure occurred. Build a simple Naive Bayes classifier and predict failure for a new record: (Vibration: High, Heat: High, Noise: Medium).

Record	Vibration	Heat	Noise	Failure
1	Low	Low	Low	No
2	High	High	High	Yes
3	Medium	Low	Medium	No
4	High	High	Medium	Yes
5	Low	Medium	Low	No

b) What assumptions does Naive Bayes make, and when can it fail? [7+3]

5. a) Apply PCA step-by-step on the following patient dataset (2D projection). Find the direction of maximum variance.

Glucose	Blood Pressure
90	60
85	65
100	70
95	68
87	63

b) Explain the difference between PCA and LDA. [7+3]

6. a) Classify the following tweet as Positive or Negative using KNN with k=3. New Tweet has Length = 18 words and #Emojis = 3.

Tweet	Length	Emojis	Sentiment
T1	20	2	Positive
T2	15	4	Negative
T3	25	3	Positive
T4	17	5	Negative
T5	19	3	Positive

b) What happens when k is too low or too high? [7+3]

7. a) Define cross-validation and explain how k-fold cross-validation works. Compare it with leave-one-out cross validation (LOOCV) in terms of bias and variance.

b) Explain how lack of explainability in black-box models can lead to ethical concerns and affect user trust in real-world AI systems. [7+3]

——————— × ———————